Energy from Waste

1Mt Carbon Management Plan for Waste-to-Energy Plants in the US

Group 4

Carbon Management Final Presentation

December 1, 2025

EAEE E4300

COLUMBIA ENGINEERING

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Introduction - Summary & Rationale

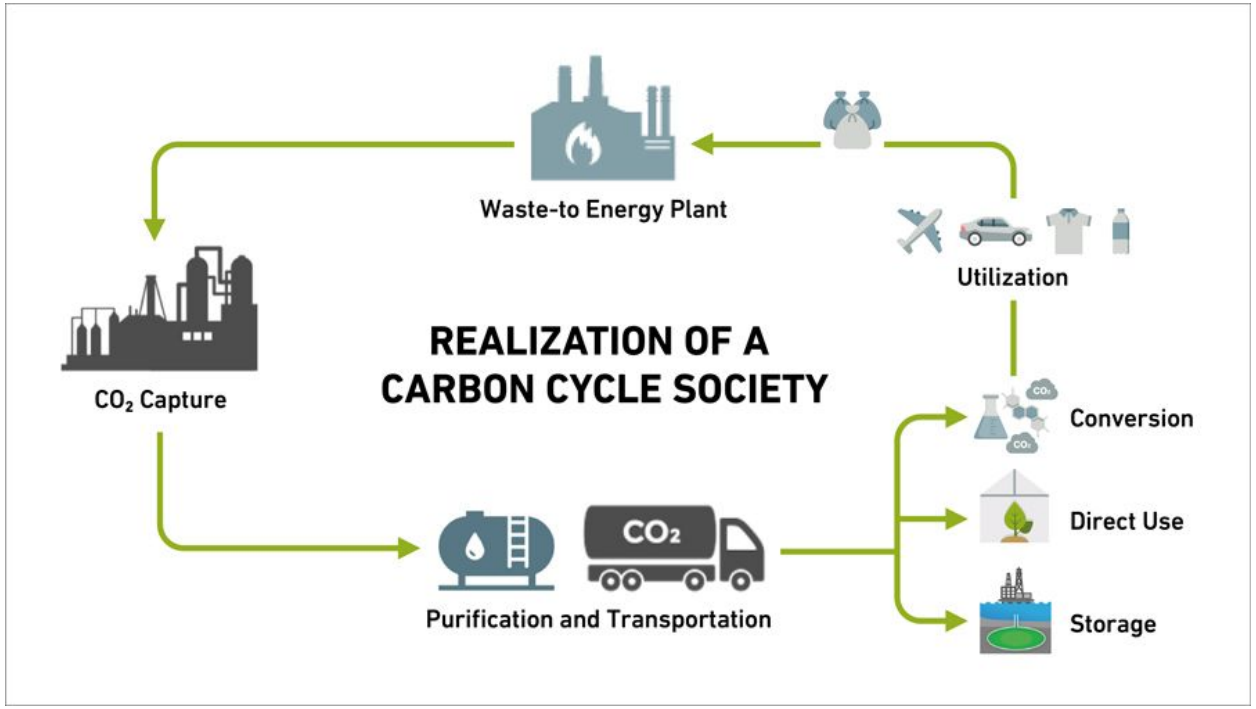


Figure 1: Cycle of CO_2 capture, utilization, and storage from WTE plant [1]

Summary

- Goal is to capture **1 Mt CO_2/yr** from U.S. WTE plants
- Use **PSC** + algae **utilization** at WWTPs + geologic **storage**
- Turns WTE into a **net-negative waste pathway**

Rationale

- Relatively cheap capture due to **high CO_2 concentration**
- WTE is a reliable source that **operates year-round**
- Close proximity to WWTPs **enables circular use**

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Introduction - Co-benefits

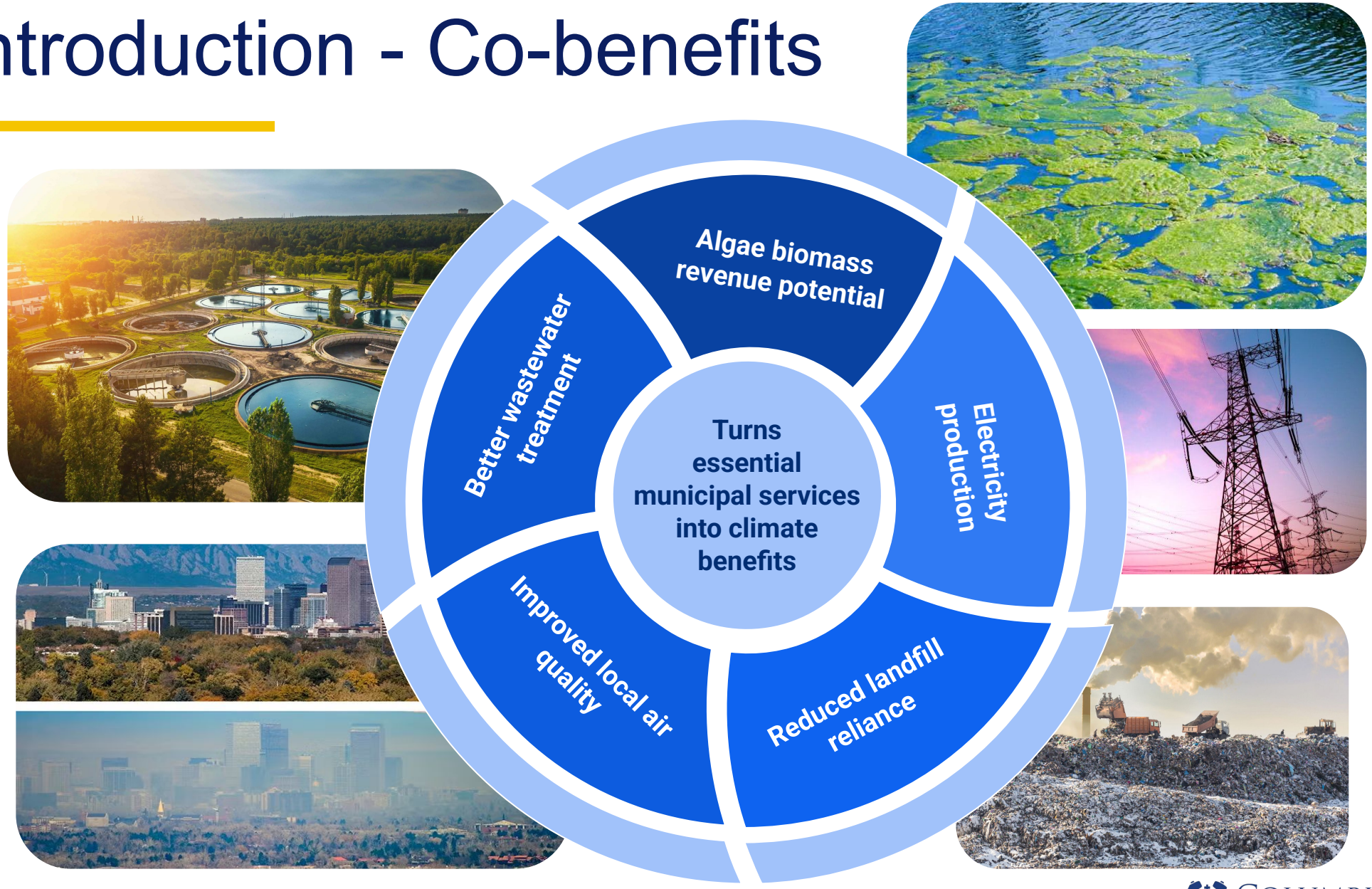
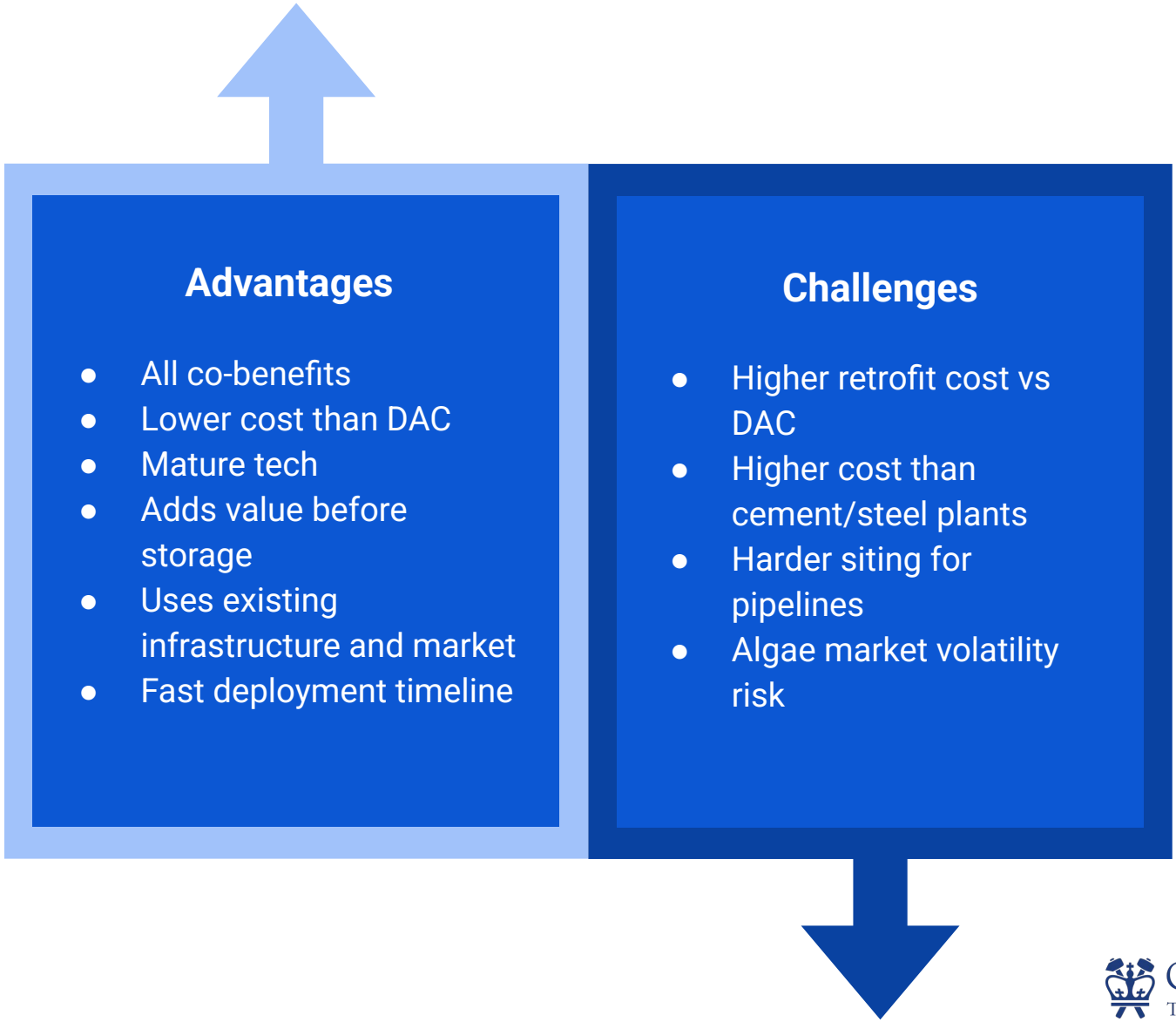


Figure 3: Co-benefits of this project

Introduction - Our method vs alternatives



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Technical Analysis

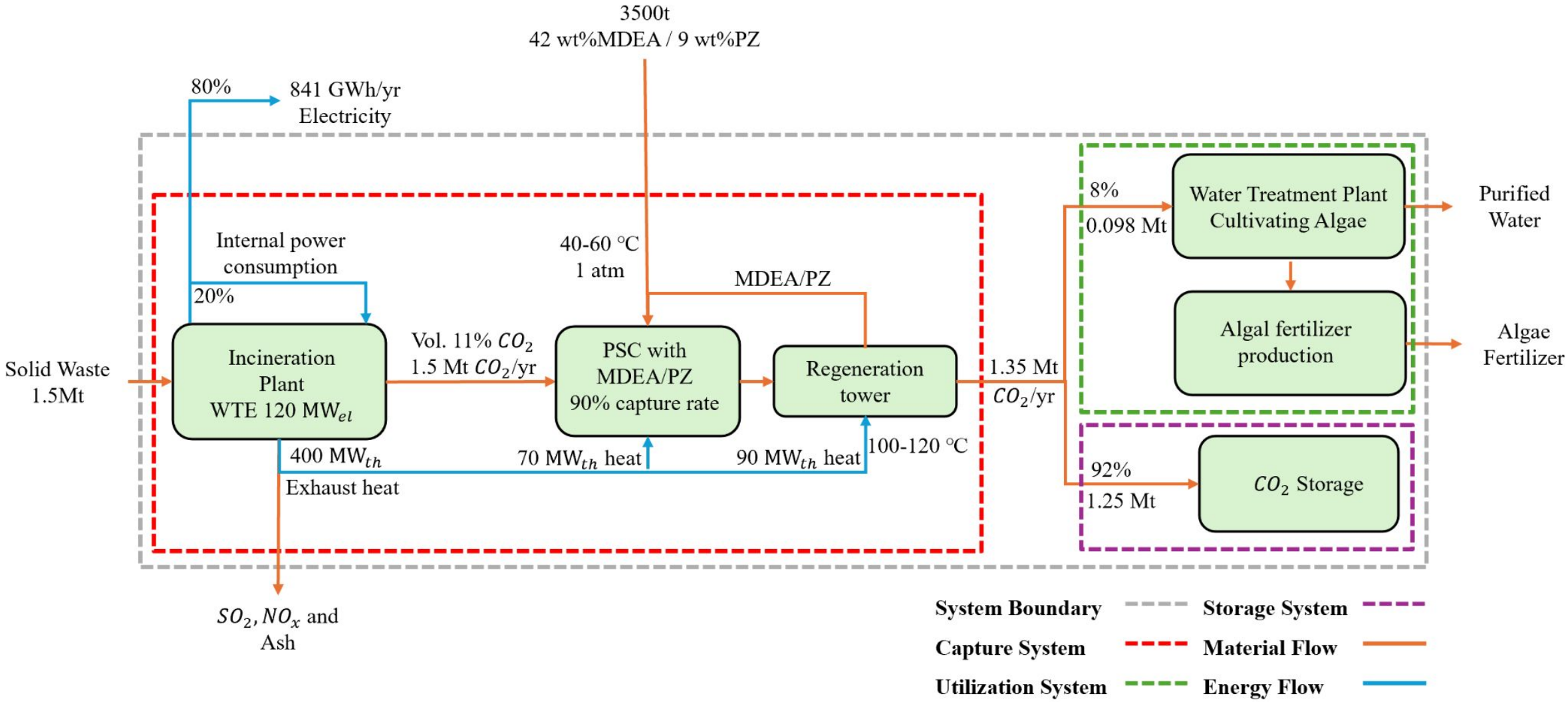


Figure 3: Flow diagram of the project

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Technical Analysis - CO_2 Capture

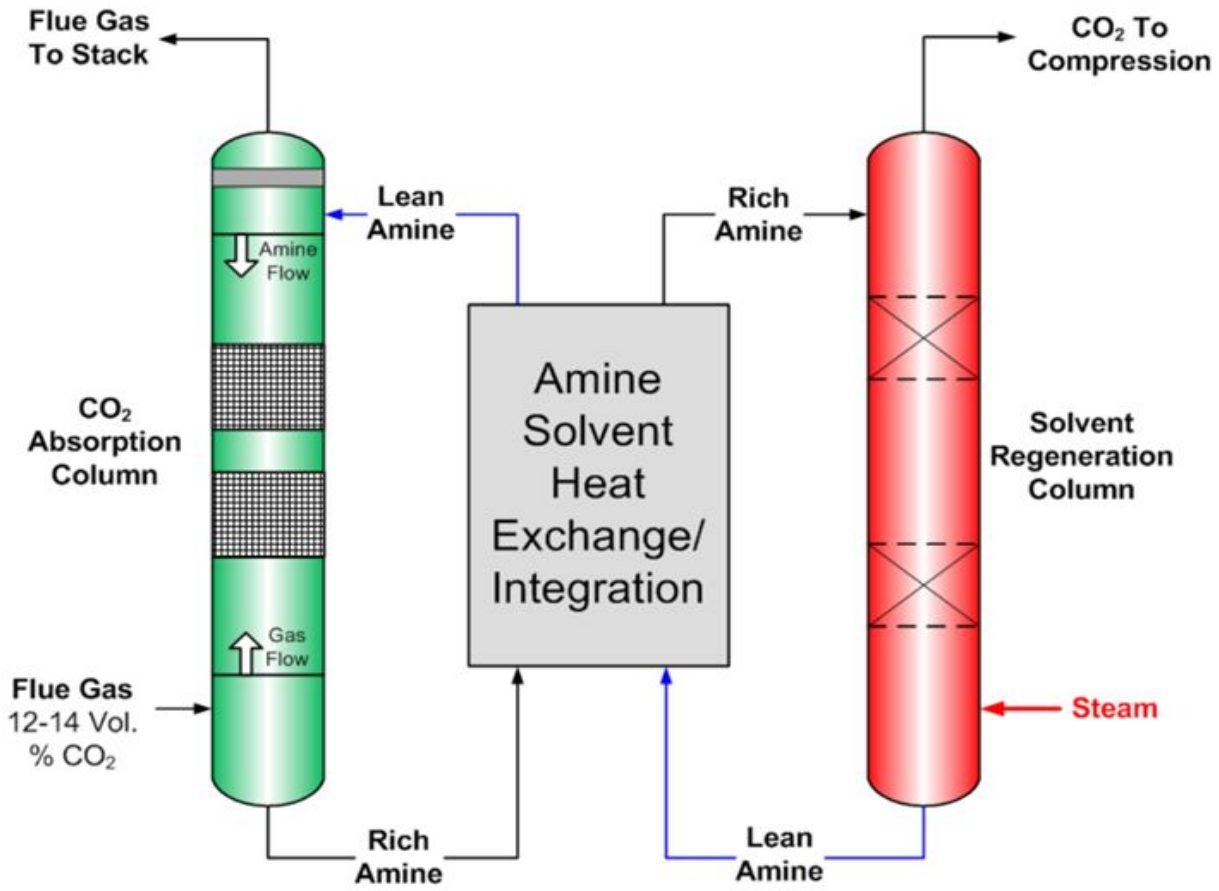


Figure 4: Capture technology

Capture (40-60℃):

$PZ + CO_2 \rightleftharpoons PZCOO^- + H^+$

$MDEA + H^+ \rightleftharpoons MDEAH^+$

PZ: Quickly capture CO_2

MDEA: Create an alkaline environment

Regeneration (100-120℃):

$PZCOO^- + H^+ \rightleftharpoons PZ + CO_2$

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Technical Analysis - CO_2 Capture

Performance Metric	MEA	MDEA/PZ	Advantage
CO_2 Working Capacity	0.50 mol CO_2/kg solvent	0.75 mol CO_2/kg solvent	MDEA/PZ is about **50% higher**
Heat of CO_2 Absorption	84 kJ/mol CO_2	75 kJ/mol CO_2	MDEA/PZ is **lower by 10–12%**, reducing regeneration energy.
Oxidative Degradation	0.39 mmol·L^{-1}·h^{-1}	0.011–0.016 mmol·L^{-1}·h^{-1}	MDEA/PZ much **longer** solvent life.
Thermal Stability	MEA degrades significantly via carbamate polymerization	MDEA/PZ stable at 100–120 °C; at 135 °C PZ begins to degrade (−44 mmolal/day)	**More thermally stable** at typical stripper temperatures (≤120 °C)
Land area requirement	About 20 000 to 40 000 m²		Same

Table 1: MEA vs MDEA/PZ. All data come from Closmann et al., 2009 [6]

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Technical Analysis - CO_2 Utilization: Algae in WWT

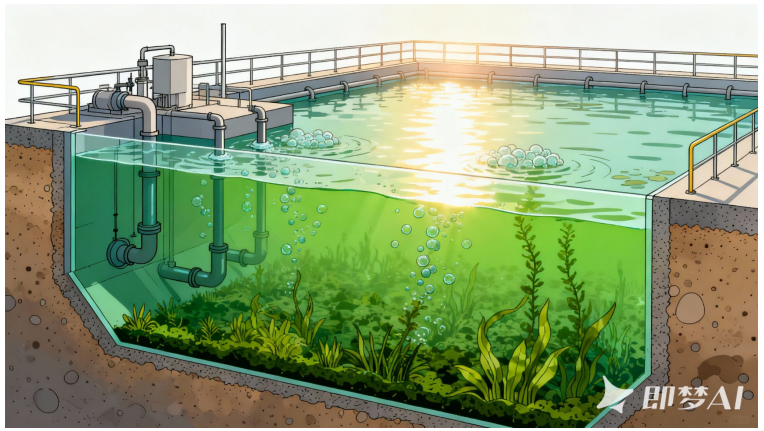




Figure 4: Algae ponds in WWT plants

- **Water treatment stage:** After sedimentation and chemical treatment, the waste water enters the tertiary polishing pond with microalgae.
- **Process:** Microalgae use CO_2, light and nutrients to produce biomass and oxygen.
- **Benefits:** Make the water more purified by — removing remaining nitrogen and phosphorus, and degrade or trap micro-pollutants, while CO_2 enhances their growth and efficiency.



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Technical Analysis - CO_2 Utilization: Algae in WWT

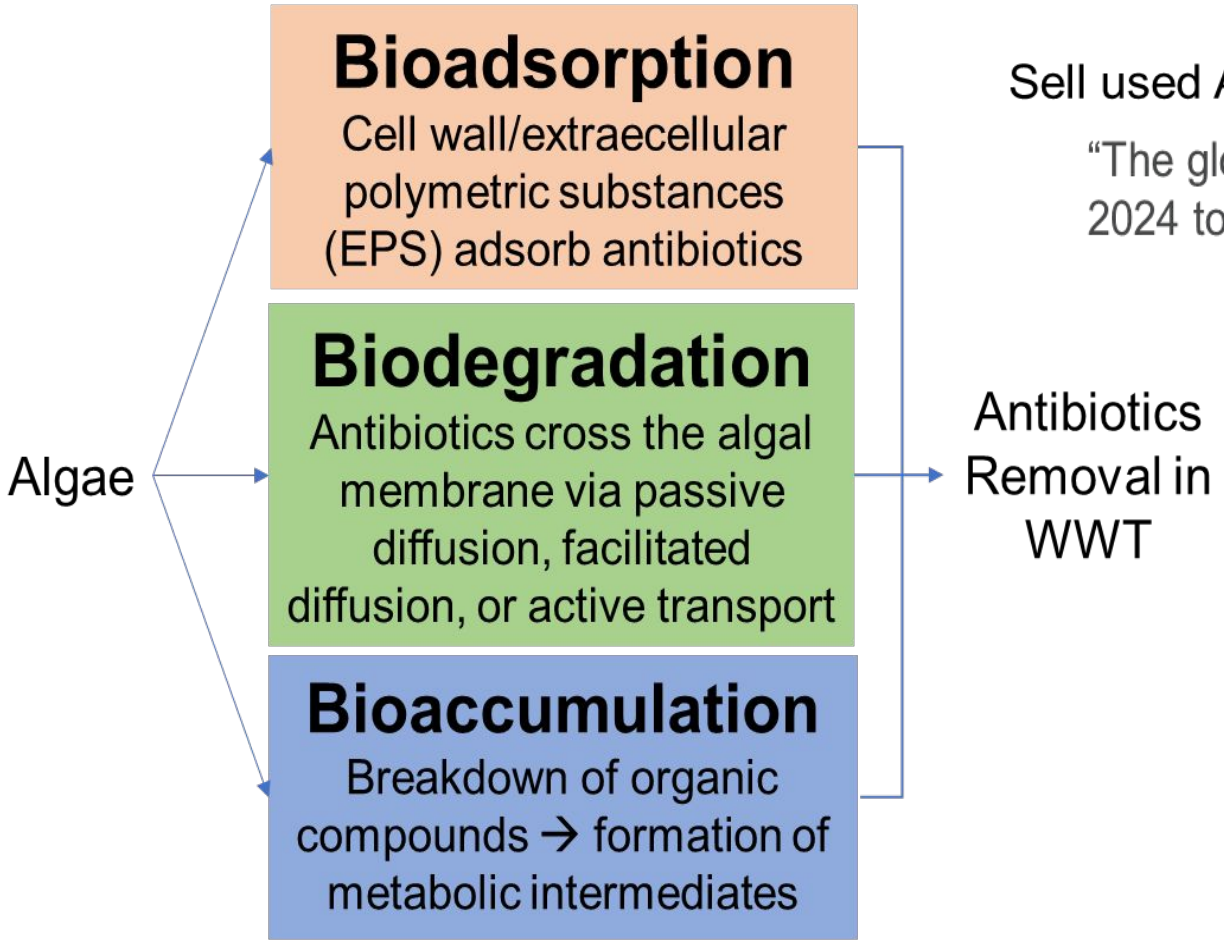


Figure 5: Antibiotics removal process in WWT plants using algae

Circular Economy

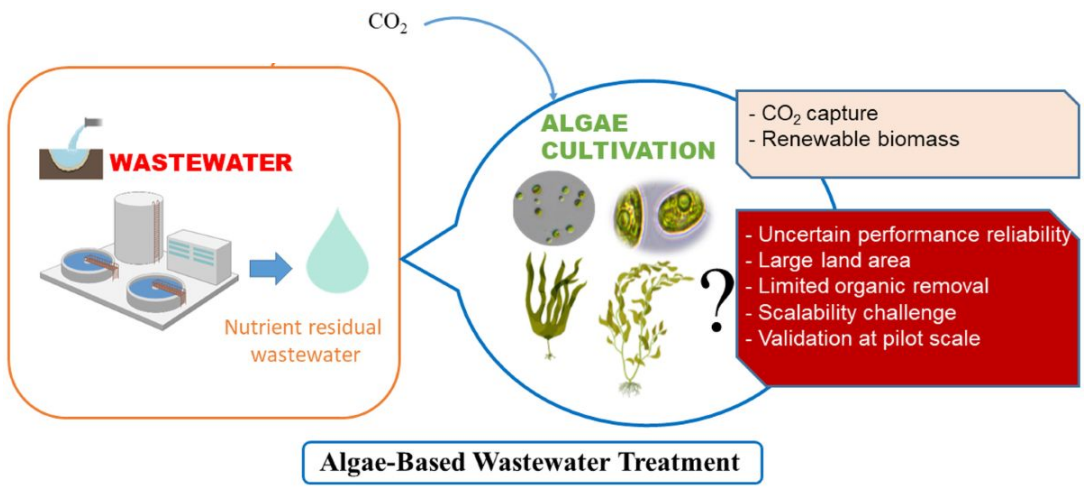
Sell used Algae → companies (Algenol Biotech) → fertilizers/biofuels

"The global algae fertilizer market grew from USD 694.86 million in 2024 to USD 748.55 million in 2025. " (Algae fertilizer market, 2025)



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Technical Analysis - CO_2 Utilization: Algae in WWT



extra algae
sell directly

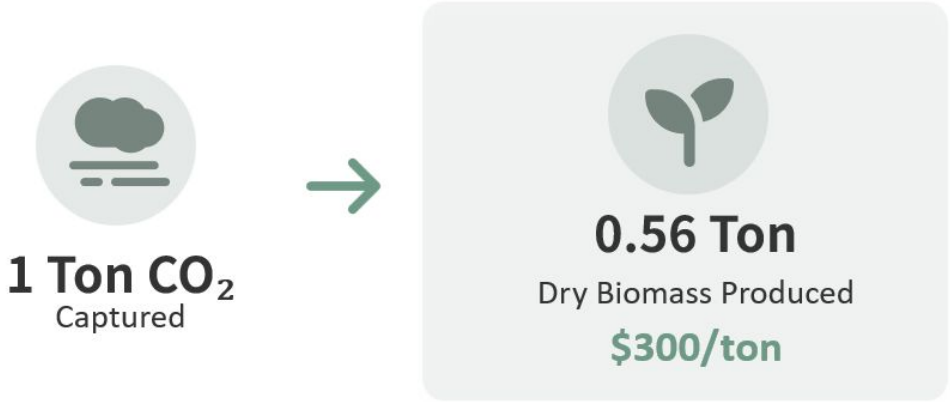


Figure 6: Collection of algae can lead to $168 revenue per ton CO_2

CO2 Utilization Estimate

Algae productivity: 0.02 kg/m²/day (dry biomass)

CO2 fixed: 50% C → 1 kg biomass ≈ 1.8 kg CO_2 fixed

Corresponding CO_2 uptake: 13.14 kg CO_2 / m² / year

Total Tertiary ponds Area: 100 * 75,000 m² /per plant

Annual CO_2 fixation: 0.098 Mt / year

So we can fix 8% of the CO2 we captured !!!

Transportation Cost (pipeline)

($20/tCO2) (0.098Mt) = $2940000

Algae initial purchase cost

($2/t) (2t) = $4000

Algae biomass revenue

($300/tCO2) (0.098Mt) = $16464000

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Technical Analysis - CO_2 Storage

Method & Trapping

Geologic storage in deep carbonate aquifers & depleted oil reservoirs. Relies on physical/geochemical trapping (structural, residual, solubility).

Injection & Containment

Supercritical CO2 injected into porous (10-20%) limestone & dolomite several kilometers deep. Containment is ensured by a sealing caprock[2].

Capacity & Regulation

Stores ~1.2 billion tons for 1000s of years. Governed by EPA Class VI Well Rule, requiring seal integrity, geochemical analysis, and long-term monitoring.

Viability & Cost

A secure sequestration location. Using existing oil reservoirs and infrastructure significantly reduces capital and maintenance costs.

Economic Analysis: Cost Breakdown

Functional Unit

1 metric ton of CO_2 captured.

Revenue Stream	WTP for cultivating Algae	16.464
	45Q Credit (Utilization)	5.88
	45Q Credit (Storage)	75.00
Total Revenue		**97.34** M

$34.5

COST PER TON CO_2

$849.63

COST PER TON OF ALGAE

CAPEX Breakdown ($118M Total)

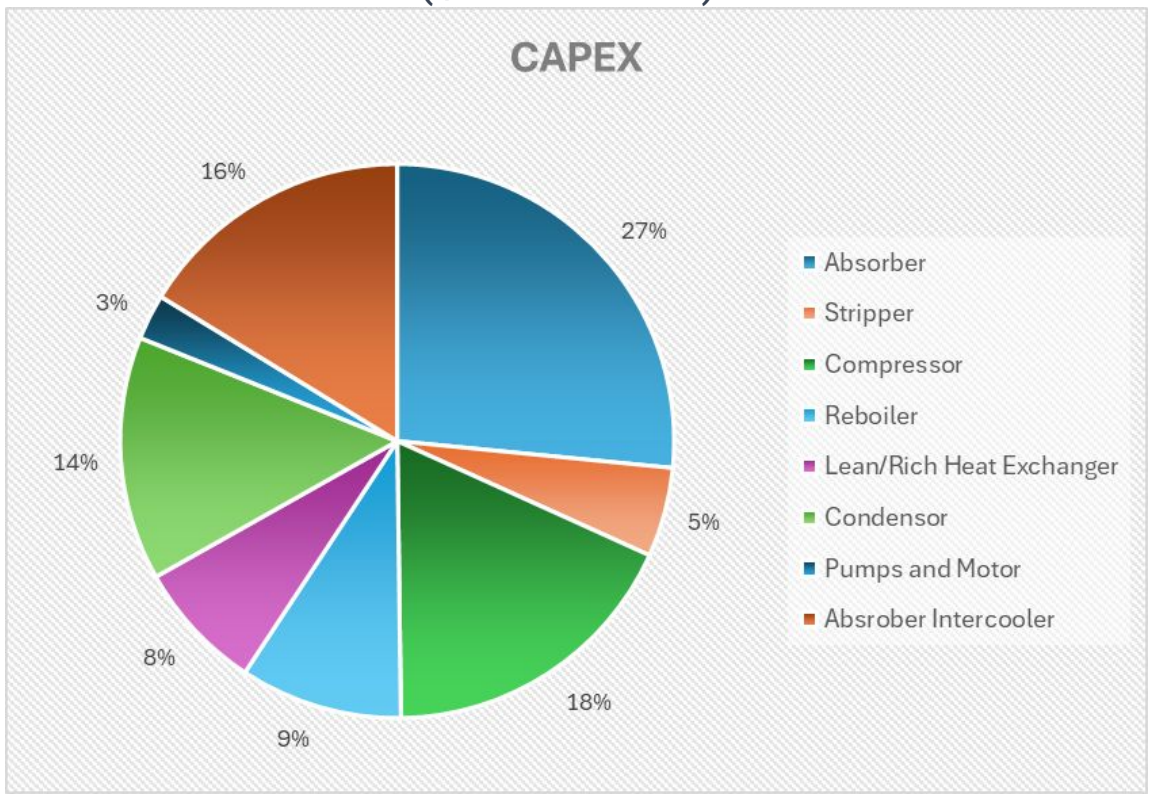


Figure 7: CAPEX Breakdown [7]

OPEX Breakdown ($35.45M annually)

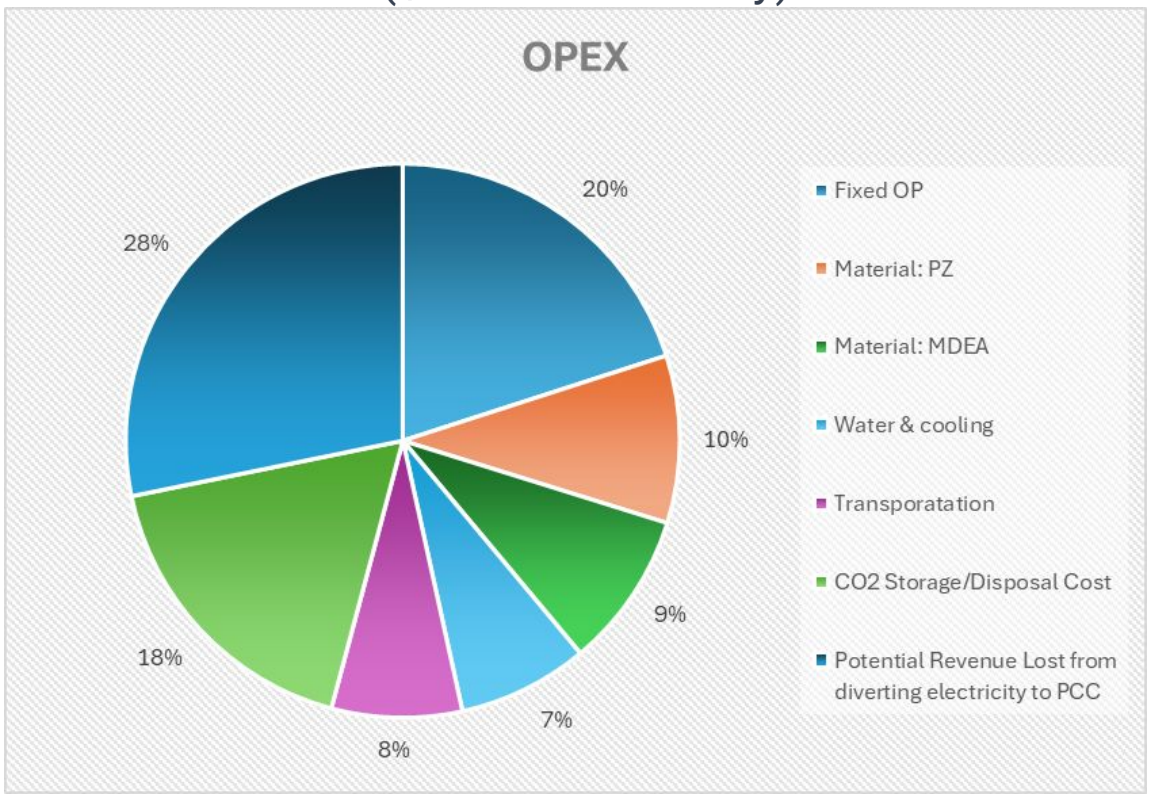


Figure 8: OPEX Breakdown [8]

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Economic Analysis: NPV & 20-Year Cash Flow Profile

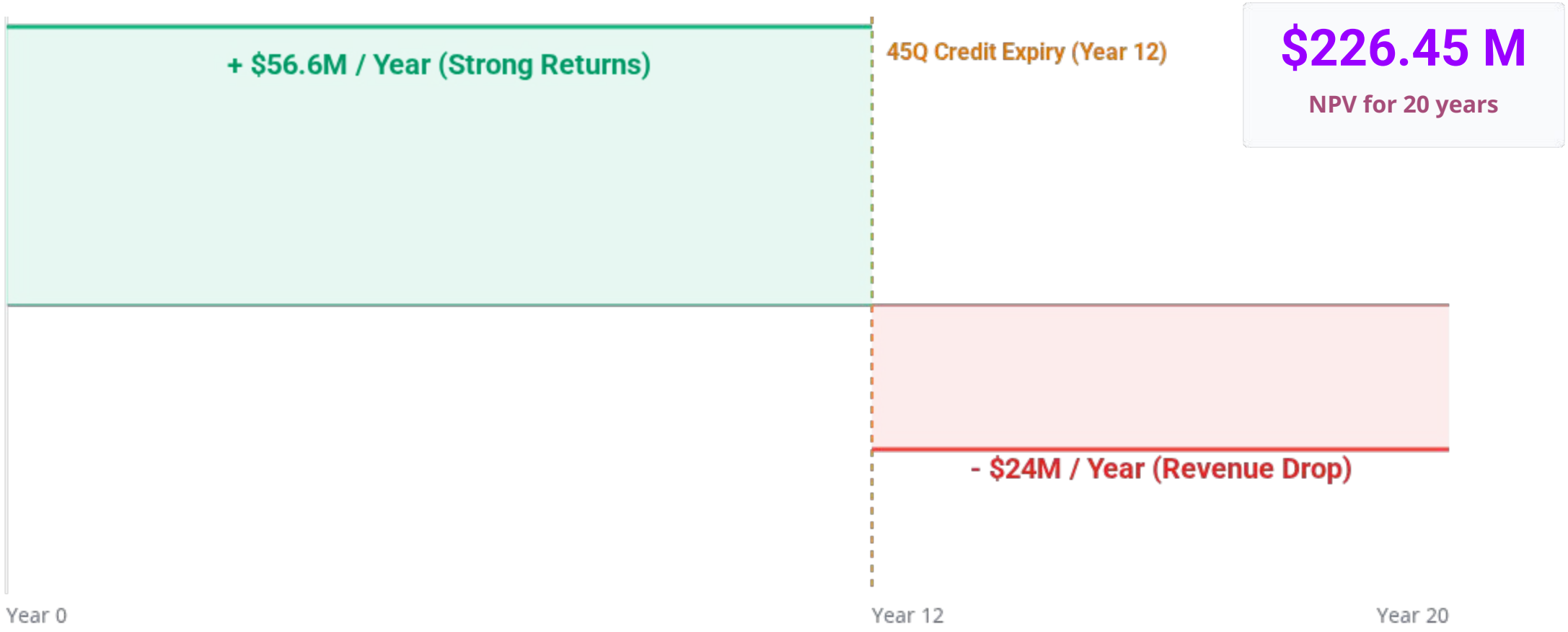
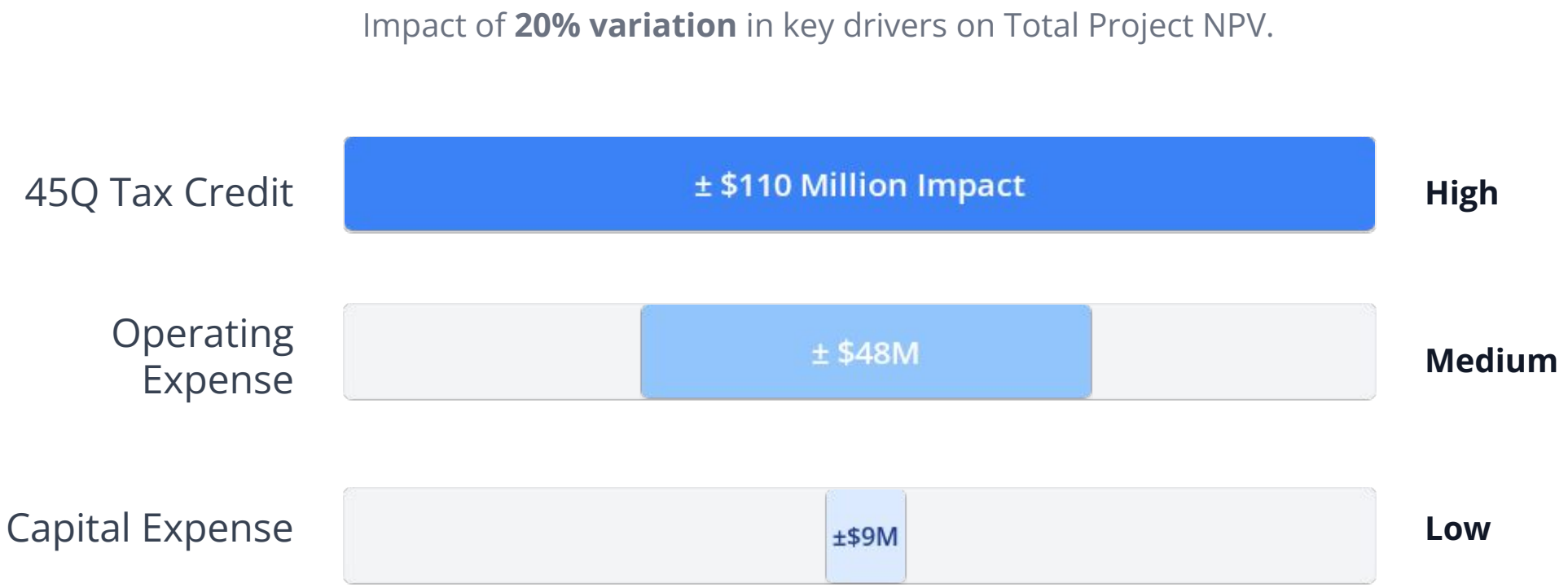


Figure 8:Cash flow profile over 20 yrs

Economic Analysis: Sensitivity Analysis (NPV Impact)



Legislative stability is the single most critical factor for success project. Also, OPEX is much more sensitive than CAPEX

Figure 9: Sensitivity analysis for NPV

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Environmental Life Cycle Analysis

- **Goal**:
  - quantify our net annual CO2 emissions from WTE + CCS system
  - compare to a conventional WTE with no carbon capture
  - functional unit: 1 ton CO2 captured from WTE flue gas and stored geologically
  - system boundary: cradle to grave: MSW -> WTE -> CO2 capture -> compression and transport -> algae-based WWTP utilization -> geologic storage
  - operational emissions only (exclude construction of pipelines, columns, and equipment)

- **CO2 sinks**: 1.0 Mt/year captured at WTE stack (90% capture), algae utilization ~8% of captured CO2, geological storage ~92% injected underground

- **CO2 sources**: residual uncaptured stack emissions, grid electricity for regeneration and compression, pipeline transport and injection energy, and algae pond operations

- **Storage Permeance**: Sunniland formation capacity ~1.2 billion tons CO2 with an expected storage lifetime of thousands to millions of years. Algae provides temporary storage and fertilizer displacement credits

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Environmental Life Cycle Analysis

- **Net CO2 performance**: gross capture of 1.0 Mt CO2/year, operational Emissions of 0.2 Mt CO2/year, net removal: 0.8 Mt CO2/year (67% reduction vs baseline)
- **Major Positive Contributors**: high capture efficiency (90-95%), majority of CO2 sent to permanent geological storage, algae CO2 uptake and avoided fertilizer production
- **Major Negative Contributors**: electricity for solvent regeneration and compression, pumping energy for pipelines and injection, algae facility operations, water and land impacts

- **Sensitivity Analysis (+/- 20-30%)**:
  - Grid CO2 intensity is the largest driver (switch to renewable)
  - A capture efficiency increase from 90-95% cuts residual stack emissions by 50 % and increases net captured CO2 by 5%
  - Algae allocation can reduce fertilizer production from more impactful sources
  - Transport distance and pressure linearly increase emissions
- **Comparison to Baseline WTE**:
  - Baseline (no capture): 1.2 Mt CO2/year (1.1 Mt/year from combustion and 0.1 Mt/year from indirect energy)
  - Our process: 1.0 Mt/ year captured, 0.2 Mt/year emitted, Net removal of 0.8 Mt/year (strongly net negative)

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Geographical Location

1. Point Source Capture: Pinellas & Hillsborough Counties WTE plants process ~1,800–2,700 tons/day each



2. Transportation: Existing pipelines for Port Tampa Bay already being studied as a CO_2 aggregation hub



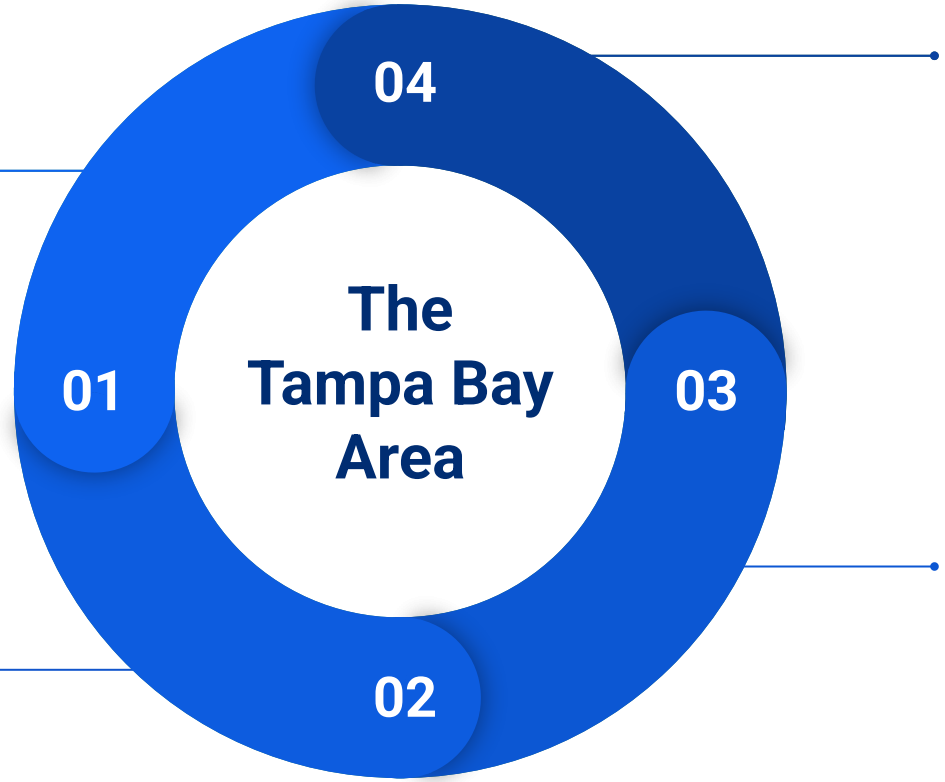
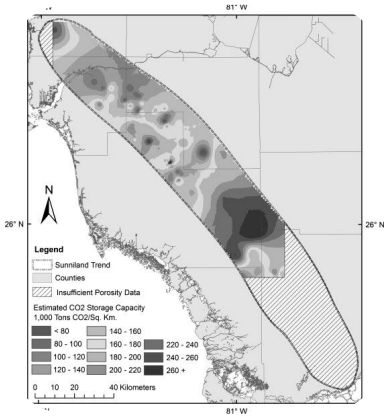


Figure 10: Co-location of process in the Tampa Bay Area

4. Storage: Sunniland Trend can store ~1BT of CO_2



3. Utilization: There are ~2000 wastewater facilities in Florida that can support algae production



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Conclusion & Recommendations

Key Points

- Technically viable
- Economically favorable with tax incentives
- Many co-benefits
- Turns essential municipal services into climate benefits

Recommendations

- Move forward with investment
- Adopt state carbon credits/tax incentives to further improve economics
- Promote credit purchases for corporate emissions offsetting.

Next Steps

- Review detailed techno-economic model
- Begin regulatory + permitting review
- Conduct site-specific geologic characterization
- Engage potential credit buyers and state agencies

Q&A



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Appendix

Potential revenue loss from electricity diversion

Table for potential revenue loss from electricity diversion				
	If Electricity Is not Diverted	After Diverting	Units	Reference
Amount for sale	1051250	841000	MWh/year	
Revenue by selling	$ 52,562,500	$ 42,050,000	USD/year	Electricity sale price = $0.05/kWh = $50/MWh
	$ 53	$ 42	MMUSD/year	
Loss	$ 10.51		MMUSD/year	

Cash Flow Sheet

*All values in MMUSD		Year 0	Year 1	Year 2	Year 3	Year 4	Year 5	Year 6	Year 7	Year 8	Year 9	Year 10	Year 11	Year 12	Year 13	Year 14	Year 15	Year 16	Year 17	Year 18	Year 19	Year 20
Equipment Capital Cost	Absorber	28																				
	Stripper	5.5																				
	Compressor	19																				
	Reboiler	10																				
	Lean/Rich Heat Exchanger	8																				
	Condensor	15																				
	Pumps and Motor	2.8																				
	Absrober Intercooler	17.16																				
Total CapEX		**105.46**																				
Installation Cost		**12.7**																				
Operating Cost	Fixed OP		7.10	7.10	7.10	7.10	7.10	7.10	7.10	7.10	7.10	7.10	7.10	7.10	7.10	7.10	7.10	7.10	7.10	7.10	7.10	7.10
	Material: PZ		3.47	3.47	3.47	3.47	3.47	3.47	3.47	3.47	3.47	3.47	3.47	3.47	3.47	3.47	3.47	3.47	3.47	3.47	3.47	3.47
	Material: MDEA		3.23	3.23	3.23	3.23	3.23	3.23	3.23	3.23	3.23	3.23	3.23	3.23	3.23	3.23	3.23	3.23	3.23	3.23	3.23	3.23
	Water & cooling		2.7	2.7	2.7	2.7	2.7	2.7	2.7	2.7	2.7	2.7	2.7	2.7	2.7	2.7	2.7	2.7	2.7	2.7	2.7	2.7
	Transporatation		2.7	2.7	2.7	2.7	2.7	2.7	2.7	2.7	2.7	2.7	2.7	2.7	2.7	2.7	2.7	2.7	2.7	2.7	2.7	2.7
	CO2 Storage/Disposal Cost		6.25	6.25	6.25	6.25	6.25	6.25	6.25	6.25	6.25	6.25	6.25	6.25	6.25	6.25	6.25	6.25	6.25	6.25	6.25	6.25
	Potential Revenue Lost from diverting electricity to PCC		10.00	10.00	10.00	10.00	10.00	10.00	10.00	10.00	10.00	10.00	10.00	10.00	10.00	10.00	10.00	10.00	10.00	10.00	10.00	10.00
Total OpEX			**35.45**	**35.45**	**35.45**	**35.45**	**35.45**	**35.45**	**35.45**	**35.45**	**35.45**	**35.45**	**35.45**	**35.45**	**35.45**	**35.45**	**35.45**	**35.45**	**35.45**	**35.45**	**35.45**	**35.45**
Maintainance Cost			**5.273**	**5.273**	**5.273**	**5.273**	**5.273**	**5.273**	**5.273**	**5.273**	**5.273**	**5.273**	**5.273**	**5.273**	**5.273**	**5.273**	**5.273**	**5.273**	**5.273**	**5.273**	**5.273**	**5.273**
Revenue Stream	WTP for cultivating Algae		16.464	16.464	16.464	16.464	16.464	16.464	16.464	16.464	16.464	16.464	16.464	16.464	16.464	16.464	16.464	16.464	16.464	16.464	16.464	16.464
	45Q Credit (Utilization)		5.88	5.88	5.88	5.88	5.88	5.88	5.88	5.88	5.88	5.88	5.88	5.88	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	45Q Credit (Storage)		75.00	75.00	75.00	75.00	75.00	75.00	75.00	75.00	75.00	75.00	75.00	75.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Total Revenue			**97.34**	**97.34**	**97.34**	**97.34**	**97.34**	**97.34**	**97.34**	**97.34**	**97.34**	**97.34**	**97.34**	**97.34**	**16.46**	**16.46**	**16.46**	**16.46**	**16.46**	**16.46**	**16.46**	**16.46**
Cash Flow in year i			**56.62**	**56.62**	**56.62**	**56.62**	**56.62**	**56.62**	**56.62**	**56.62**	**56.62**	**56.62**	**56.62**	**56.62**	**-24.26**	**-24.26**	**-24.26**	**-24.26**	**-24.26**	**-24.26**	**-24.26**	**-24.26**

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Calculation of NPV

Year	Net Profit (Cash Flow)	PWF	NPV
0	-118.1152	1	-118.12
1	56.62	0.990099	51.47
2	56.62	0.980296	46.80
3	56.62	0.97059	42.54
4	56.62	0.96098	38.67
5	56.62	0.951466	35.16
6	56.62	0.942045	31.96
7	56.62	0.932718	29.06
8	56.62	0.923483	26.41
9	56.62	0.91434	24.01
10	56.62	0.905287	21.83
11	56.62	0.896324	19.85
12	56.62	0.887449	18.04
13	-24.26	0.878663	-7.03
14	-24.26	0.869963	-6.39
15	-24.26	0.861349	-5.81
16	-24.26	0.852821	-5.28
17	-24.26	0.844377	-4.80
18	-24.26	0.836017	-4.36
19	-24.26	0.82774	-3.97
20	-24.26	0.819544	-3.61
	Total NPV (MMUSD)		226.45

45Q change	NPV Value	Net Change (MMUSD)
20%	336.67	110.22
0	226.45	
-20%	116.24	110.22

CAPEX change	NPV Value	Net Change (MMUSD)
20%	217.48	-8.97
0	226.45	
-20%	235.43	-8.98

OPEX Change	NPV Value	Net Change (MMUSD)
20%	178.15	-48.30
0	226.45	
-20%	274.76	-48.31

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